

Welcome to A Catholic Introduction to Artificial Intelligence Class 9! AI and the Future of Work

1. Overview

Of all the questions raised by artificial intelligence, none is more personal or more urgent than this: what will AI do to work? Work is not simply how people earn money. It is how people contribute to society, develop their skills, provide for their families, and find purpose and dignity in daily life. When a technology arrives that can perform an enormous range of tasks faster, cheaper, and more consistently than human beings, the consequences extend far beyond economics.

This class examines those consequences honestly. We look at which kinds of work AI is already replacing, which kinds it is augmenting, and which entirely new categories of work it is creating. We analyze what economists, researchers, and institutions have predicted about the future labor market, and we consider what those predictions get right and what they might be missing. We then turn to Catholic social teaching, which has a rich and specific tradition of thinking about the dignity of work, the rights of workers, and the obligations of societies when technology disrupts established ways of life. We will also look at Pope Leo XIV's encyclical on artificial intelligence called *Magnifica Humanitas*.

The class ends with a structured debate on the central question: will AI ultimately create more jobs than it eliminates?

2. Learning Objectives

By the end of this class, students should be able to:

1. Distinguish between jobs AI is likely to replace, jobs it is likely to augment, and jobs it is likely to create.
2. Explain the concept of automation risk and identify factors that make a job more or less susceptible to it.
3. Summarize key findings from economic research on AI and employment.
4. Identify specific emerging careers in the AI economy and explain what they involve.
5. Explain Catholic social teaching on the dignity of work and apply it to questions about AI-driven automation.
6. Construct and defend an argument on both sides of the debate about AI and net job creation.

3. Class 8 Review

Class 8 explored how artificial intelligence is used in entertainment and media. We examined six major application areas: streaming recommendation systems, AI-generated music and art, deepfakes and synthetic media, AI in film and TV production, social media algorithms, and AI in journalism.

A central theme was the difference between what recommendation systems are designed to optimize — engagement — and what actually serves users well. Engagement and quality are not the same thing, and systems optimized for the former can systematically surface content that is emotionally provocative rather than genuinely valuable. The filter bubble problem showed how algorithmic personalization can narrow exposure to different perspectives over time.

On AI-generated creative work, we considered questions of authorship, consent, and economic justice — particularly the use of human-created work to train models that then compete with those same human creators. Deepfakes introduced the most direct threat to truth: AI capable of fabricating convincing video and audio of real people saying things they never said, with implications for political disinformation, non-consensual imagery, and financial fraud.

Catholic teaching on truth, human dignity, and the common good provided the evaluative framework for each application. The Eighth Commandment's call to bear true witness, the concept of imago Dei, and the principle of subsidiarity all apply directly to how media technology should be developed and used

4. What Work Is and Why It Matters

It is tempting to reduce questions about AI and work to questions about employment — specifically, how many jobs will exist and at what wages. But this framing misses something

important. Work, in the fullest sense, is one of the primary ways human beings express their capabilities, contribute to their communities, and find meaning in their lives.

A parent raising children is doing work, even without a paycheck. A volunteer building homes is doing work. An artist perfecting a craft is doing work. The economic dimension of work — the fact that most people need to earn income to survive — is real and urgent, but it does not exhaust what work means. When we ask what AI will do to work, we need to hold both dimensions in view: the economic and the human.

The Three Functions of Work

Economists and social thinkers have identified three distinct functions that work serves in human life: **What does Catholic Social Doctrine say?**

- Economic function — work generates income that allows people to meet their material needs and those of their families
- Social function — work provides structure, community, identity, and a sense of belonging and contribution
- Personal function — work develops skills, expresses creativity, builds character, and provides a sense of purpose and accomplishment

AI automation that eliminates jobs threatens all three functions simultaneously. A person who loses their job does not only lose income. They may also lose the social connections of the workplace, the daily structure that organizes their time, and the sense of purpose that comes from contributing skilled effort to a meaningful task. These losses are real even when income is replaced through other means, such as a universal basic income.

4.1 Jobs AI Is Replacing

Not all jobs are equally exposed to automation. Researchers have developed frameworks for assessing which tasks — and therefore which jobs — AI is most likely to replace. An early influential early framework, developed by economists Carl Benedikt Frey and Michael Osborne at Oxford in 2013, estimated that approximately `47%` of US jobs were at high risk of automation within two decades. A later study (OECD, 2018; Brookings, 2019) significantly lowered that rate to `10–25%` of US jobs being at high risk of automation.

The key insight is that automation risk is determined by task structure, not job title. Jobs that consist primarily of well-defined, repetitive tasks with predictable inputs and clear outputs are highly automatable. Jobs that require open-ended judgment, physical dexterity in unpredictable environments, social intelligence, or creative problem-solving are much harder to automate. Even with highly automatable jobs, there needs to be a `human-in-the-loop` (HIL).

Tasks most vulnerable to AI replacement share several features:

- Routine and repetitive — the same steps performed in the same order, day after day

- Rule-based — decisions that can be reduced to explicit if-then logic
- Data-intensive but judgment-light — processing large volumes of structured information without needing to interpret context
- Predictable environments — physical or information environments that do not change unexpectedly
- High volume, low variability — the same task performed thousands of times with minor variations

This profile describes a wide range of current jobs:

Manufacturing and assembly: Repetitive physical tasks on factory floors — attaching components, inspecting products, packaging — are being automated by robotic systems. AI-guided robots can now perform assembly tasks that previously required human hands, working continuously without fatigue. However, small to medium size companies will have difficulty allocating budgets to implement AI automation. AI automation, like all automation, comes with investment, implementation, and maintenance costs.

Data entry and processing: Entering information into systems, extracting data from documents, categorizing records, and generating routine reports are being replaced by AI tools that can process documents faster and with greater accuracy than human clerks.

Basic customer service: Call centers and customer support roles focused on answering standard queries, processing returns, or providing product information are being replaced by AI chatbots and virtual assistants that can handle thousands of simultaneous interactions. However, who doesn't want to talk/chat with a live agent when circumstances are complex and nuanced?

Routine legal and financial work: Document review, contract analysis, basic due diligence, and standard financial reporting can now be performed by AI tools faster and more cheaply than by junior lawyers or analysts. Large law firms have already reduced their associate intake as a result.

Transportation: Long-haul trucking, taxi and ride-share driving, and warehouse logistics are targets for automation through self-driving vehicles and autonomous robots. The transition is much slower than early predictions suggested as consumers and investors express safety concerns.

A critical question is not just which jobs will be automated, but how quickly. Previous waves of automation — agricultural mechanization in the nineteenth century, industrial automation in the twentieth — displaced workers over **decades**, allowing time for economies and social systems to adapt. AI-driven automation `may` (but not likely) move much faster, compressing timelines that previously allowed for gradual adjustment.

This speed matters enormously for policy and investors. A transition that takes fifty years allows workers to retrain and for educational systems to adjust. A transition that takes ten

years may not. Investors want to see gains in their lifetime.

4.2 Jobs AI Augment

Augmentation vs. Replacement

Not every job that AI touches is a job that AI replaces. In many fields, AI functions as a powerful tool that makes human workers more productive, more capable, and more effective — without replacing them. This is called augmentation: the human and the AI work together, each contributing what the other cannot. This is not different when Microsoft Office was first released. Users went from handwritten ledgers and typed memos to Excel and Word. Employees needed to adapt. Employees that couldn't adapt were either reassigned or forced to find a new job.

The general pattern is that AI handles the parts of a job that are data-intensive, repetitive, or require processing speed beyond human capacity — freeing the human worker to focus on the parts that require judgment, creativity, relationship, and moral accountability.

Examples of Augmented Professions

Medicine and healthcare: AI can analyze medical images faster and with accuracy that matches or exceeds specialist radiologists. But the radiologist does not disappear — they interpret the AI's findings in the context of a patient's full history, communicate with patients and families, make judgment calls the algorithm cannot, and bear moral and legal responsibility for decisions. AI makes the radiologist more capable; it does not replace them.

Engineering and architecture: AI tools can generate design options, run simulations, identify structural weaknesses, and optimize for cost or energy efficiency at a speed no human team could match. But the engineer still decides what to build, why to build it, how to navigate regulatory and community concerns, and how to handle the unexpected problems that arise during construction.

Teaching and education: AI can personalize learning materials to individual students, identify which students are struggling, generate practice exercises, and provide instant feedback on written work. But the teacher's role — building relationships with students, motivating them, helping them navigate failure, serving as a moral model — is not replicated by any algorithm.

Creative fields: Writers, designers, musicians, and filmmakers increasingly use AI tools to generate drafts, explore variations, and automate production tasks. The creative judgment — deciding what is good, what is true, what serves the audience — remains human. AI is a powerful assistant; it is not an artist.

Law and consulting: Senior lawyers and consultants whose work involves strategic judgment, client relationships, and navigating ambiguity are augmented rather than replaced. AI

handles research, document review, and routine analysis; the experienced professional handles everything that requires wisdom, context, and accountability.

One consistent pattern in economic research is that augmentation tends to benefit workers with higher levels of skill and education more than those with lower levels. `AI tools are most useful when the worker using them already has a strong foundation of domain knowledge and can critically evaluate the AI's outputs.` A senior doctor using AI diagnostic tools is more capable than before. A junior doctor with gaps in their training may become dependent on AI outputs they cannot adequately evaluate.

This creates a risk: AI may widen existing inequalities in the labor market, providing the greatest benefits to already-advantaged workers while offering less protection to those in more vulnerable positions.

4.3 Jobs AI Creates

Every major technological transition in history has destroyed some jobs and created others — often jobs that no one predicted in advance. The Industrial Revolution eliminated hand-weavers and created factory managers, electrical engineers, and railroad conductors. The digital revolution eliminated typographers and travel agents and created web developers, data scientists, and social media managers.

AI will follow the same pattern. Some of the jobs it creates are already visible. Others will emerge in ways that are difficult to predict from our current vantage point. What is clear is that the AI economy will require significant human labor to build, operate, evaluate, and govern.

Emerging AI-Era Careers

Prompt engineers: Specialists who design, test, and optimize the instructions given to AI systems to produce desired outputs. This requires deep knowledge of how language models work, what their failure modes are, and how to communicate with them effectively. Already a well-compensated specialist role in technology companies.

AI trainers and data annotators: AI models require vast quantities of labeled training data. Human workers evaluate model outputs, correct errors, flag harmful content, and provide the feedback signals that shape model behavior through reinforcement learning. This work requires judgment, cultural knowledge, and ethical sensitivity that cannot be automated.

AI auditors and evaluators: As AI systems are deployed in high-stakes contexts — credit decisions, medical diagnosis, criminal justice, hiring — there is growing demand for professionals who can evaluate those systems for bias, accuracy, and compliance with legal and ethical standards. This is a new professional field with no direct historical precedent.

AI ethicists and policy specialists: Governments, corporations, and civil society organizations need people who understand both the technical capabilities of AI and the ethical, legal, and

social frameworks needed to govern them. This role combines technical literacy with philosophy, law, and policy expertise.

Human-AI interaction designers: As AI interfaces become ubiquitous, there is growing demand for designers who specialize in how human beings interact with AI systems — making those interactions intuitive, transparent, and trustworthy.

AI maintenance and operations: Deployed AI systems require ongoing monitoring, retraining, debugging, and infrastructure management. The engineers, data scientists, and operations specialists who keep these systems running constitute a large and growing workforce.

The existence of new AI-era jobs does not automatically resolve the displacement problem. The new jobs may not be located where the old jobs were. They may require significantly different skills. They may be accessible only to workers with higher levels of education. And they may not exist in sufficient numbers to absorb all the workers displaced by automation.

History suggests that technological transitions eventually produce `net employment gains` — but the transition period can be long, painful, and unequally distributed. The agricultural laborers displaced by mechanization in the nineteenth century did not all become factory workers overnight. Many experienced decades of hardship. We should not assume that the transition from an AI-impacted economy to a new equilibrium will be smooth or swift.

5. What Research Says

A substantial body of economic research has accumulated on AI, automation, and employment. The findings are more complex and contested than either optimistic or pessimistic headlines suggest. Several themes recur across the most rigorous studies:

- Task displacement is real and already measurable. Studies examining labor market data show that occupations with high proportions of automatable tasks have already seen slower wage growth and employment decline relative to occupations requiring more judgment and social skill. This pattern began with earlier waves of computerization and is accelerating.

- New job creation has historically offset displacement — but may not this time. Economists including Daron Acemoglu at MIT have argued that previous technological transitions created new tasks for humans faster than they destroyed old ones. Acemoglu's concern about AI specifically is that current AI development is heavily weighted toward automation of existing tasks, with comparatively less creation of new human-complementary tasks.

- The distribution of benefits and losses is unequal. Research consistently finds that the gains from AI-driven productivity improvements are concentrated among capital owners and highly skilled workers, while the costs of displacement are concentrated among

lower-skilled workers with fewer options. This is not an inevitable feature of the technology; it is a consequence of policy choices about taxation, redistribution, and labor market regulation.

- Geography matters. Automation impacts are not evenly distributed across regions. Communities whose economies depend heavily on automatable industries face concentrated and severe impacts. Workers in those communities often lack the geographic or financial mobility to access new opportunities elsewhere.

Honest engagement with the research requires acknowledging its limits. Economic models that predict job losses from automation often underestimate the creation of jobs in sectors that are difficult to predict from current data. The history of technological forecasting is one of consistent underestimation of the new possibilities that emerge.

What we can say with confidence is that the transition will be uneven, that some workers and communities will bear disproportionate costs, and that the outcome is not determined by the technology itself — it is determined by the social and political choices we make in response to it.

6. A Catholic Perspective

Laborem Exercens: The Dignity of Work

In 1981, Pope John Paul II published Laborem Exercens (On Human Work), one of the most important Catholic social teaching documents of the twentieth century. Written against the backdrop of industrial automation, it established principles that apply with full force to AI-driven automation today.

The central teaching of Laborem Exercens is that work has priority over capital. This means that economic systems, including technological systems, must be evaluated by how well they serve the human beings who work within them — not by how efficiently they generate profit for those who own the machines. When technology is used to eliminate work without regard for the human consequences, it violates this principle.

The Principle of the Priority of Labor

John Paul II distinguished between the objective dimension of work — what is produced — and the subjective dimension — who is doing the producing and what it means to them. The objective dimension is what technology can most easily replicate: a machine can produce the same output as a human worker, often faster and more consistently. But the subjective dimension — the fact that work is done by a person, and that doing it shapes and expresses that person's dignity — is not replicated by automation.

This is not a sentimental argument. It is a claim about the proper ordering of economic life. An economy that eliminates human work in the name of efficiency has confused means with

ends. Efficiency is a means. The development and dignity of human persons is the end.

Solidarity and the Preferential Option for the Poor

Catholic social teaching insists on solidarity: the recognition that we are responsible for one another, that the suffering of those who are displaced by automation is not merely their private misfortune but a shared social concern. This solidarity has a specific directional bias: the preferential option for the poor, which holds that social and economic arrangements must be evaluated especially by how they affect the most vulnerable.

Applied to AI and work, this means that the question is not only whether AI creates net economic growth — it is whether that growth reaches those who are most at risk. A society in which AI dramatically increases aggregate wealth while leaving displaced workers without income, purpose, or dignity has failed the test of solidarity, regardless of its GDP figures.

The Right to Work and the Duty to Provide It

The Catholic tradition holds that work is a right — not merely a market outcome, but something to which persons are entitled as a matter of human dignity. From this follows a corresponding duty on the part of societies and institutions: to structure economic life in ways that make work available and accessible to all who seek it, and to support those who cannot find it through no fault of their own.

This does not mean that automation is wrong. Technology has always changed the nature of work, and the Church has never taught that existing jobs must be preserved regardless of circumstances. What it does mean is that those who benefit from automation — business owners, investors, consumers — bear obligations to those whose livelihoods it displaces. Those obligations include supporting retraining, advocating for social safety nets, and ensuring that the gains from productivity are shared broadly rather than captured exclusively by those at the top.

Magnifica Humanitas

On May 25th, Pope Leo XIV releases his first encyclical called *Magnifica Humanitas*....

7. The Debate: Will AI Create More Jobs Than It Eliminates?

This is one of the most actively contested questions in economics and public policy today. It is not a question with a settled answer — the evidence points in multiple directions, and the outcome depends heavily on choices that societies have not yet made. For precisely this reason, it is an excellent question for structured debate.

Both sides of the debate have serious arguments supported by credible research and historical evidence. Understanding both sides well is more valuable than choosing one

prematurely.

AI will create more jobs than it eliminates	AI will eliminate more jobs than it creates
Historical precedent supports net job creation. Every previous major technological transition — mechanization, electrification, computerization — ultimately created more jobs than it destroyed, raising living standards and generating new industries. AI is likely to follow the same pattern. New industries require workers to build, operate, maintain, and govern them. Productivity gains create wealth that generates demand for new goods and services, which in turn creates jobs. The pessimists have been wrong before.	This time may be different. Previous automation replaced physical labor; AI replaces cognitive labor. There is no obvious new domain of human comparative advantage to move into when AI can match human performance across a wide range of intellectual tasks. The speed of AI development may not allow time for the gradual retraining and reallocation that smoothed previous transitions. And the concentration of AI's benefits among capital owners and highly skilled workers may mean that gains never reach displaced workers in sufficient quantity.

7.1 Arguments for Net Job Creation

- Historical pattern: automation has always produced net job growth over the long run
- New industries: AI creates entirely new sectors requiring human workers — AI development, governance, maintenance, and adjacent services
- Productivity dividend: efficiency gains lower prices, raise real wages, and generate demand that creates jobs elsewhere
- Comparative advantage: humans retain advantages in creativity, judgment, relationships, and moral accountability that AI cannot replicate
- Augmentation dominates replacement: in most professions, AI is more likely to make workers more productive than to replace them entirely

7.2 Arguments for Net Job Elimination

- Breadth of cognitive automation: unlike previous automation, AI can perform intellectual tasks across almost every domain, leaving fewer clear areas of human refuge
- Speed of transition: AI capabilities are advancing faster than educational and social systems can adapt
- Concentration of gains: even if aggregate wealth increases, there is no mechanism ensuring displaced workers benefit
- Structural unemployment: new AI-era jobs require different skills, locations, and education levels than the jobs being eliminated
- Capital substitution: AI shifts production from labor to capital, reducing the share of economic output that flows to workers

7.3 What a Catholic Perspective Adds

Catholic social teaching does not resolve this empirical debate — it reframes it. Whether AI creates or eliminates net jobs is an important question, but it is not the only important

question. Even if AI creates more jobs in aggregate, a Catholic evaluation asks:

- Are the new jobs accessible to the workers who lose the old ones?
- Do the new jobs provide the same dignity, community, and meaning as the jobs they replace?
- Are the gains from productivity shared broadly or concentrated narrowly?
- What happens to those who cannot adapt — and who is responsible for them?

A society that answers these questions poorly has failed even if its employment statistics look healthy. The measure of an economy is not its aggregate output but its service to the common good — and the common good includes those who are most vulnerable to the costs of transition.

8. Lesson Summary

This lesson examined AI's impact on work across three categories — jobs being replaced, jobs being augmented, and jobs being created — alongside the economic research on these trends and the Catholic social teaching framework for evaluating them.

Concept	What It Means	Example
Jobs AI replaces	Routine, repetitive, rules-based tasks with predictable inputs and outputs	Assembly line work, data entry, basic customer service, routine legal document review
Jobs AI augments	Complex work requiring judgment, creativity, or human relationships — AI assists but does not replace	Engineering, medicine, teaching, design, journalism
Jobs AI creates	Entirely new roles that emerge to build, maintain, govern, and work alongside AI systems	Prompt engineers, AI auditors, machine learning trainers, AI ethicists
Transition risk	Workers in automatable roles without access to reskilling or alternative employment	Truck drivers, radiologists, paralegals, call center staff
Catholic response	Reaffirm the dignity of work; ensure technology serves people; support workers displaced by automation	Retraining programs, social safety nets, preferential option for vulnerable workers

Key Takeaways

- Automation risk is determined by task structure, not job title — routine, rule-based, predictable work is most vulnerable
- AI augments many professions rather than replacing them entirely, but augmentation tends to benefit higher-skilled workers more than lower-skilled ones
- New AI-era careers are emerging, but they require different skills and locations than the jobs being displaced — making transition difficult for many workers

- Economic research shows that the costs of displacement are concentrated among the most vulnerable workers while the benefits are concentrated among owners and the highly skilled
- Catholic social teaching holds that work has priority over capital, that displaced workers have a claim on social solidarity, and that the common good — not aggregate GDP — is the proper measure of economic success
- Whether AI creates or eliminates net jobs matters — but so does who gets the new jobs, what those jobs are like, and what happens to those who cannot access them

8. Exit Ticket

AI's impact on work is not a single event but an ongoing process that will unfold differently across industries, regions, and skill levels. The outcome is not determined by the technology — it is determined by the choices societies make in response to it. Those choices are moral choices as much as economic ones.

9. Vocabulary

Term	Definition
Automation	The use of technology, machinery, or software to perform tasks previously done by human workers, reducing the need for manual or cognitive labor
Automation risk	The likelihood that a given job or task will be taken over by automated systems, typically assessed by how routine, repetitive, or rules-based the work is
Task displacement	The process by which automation eliminates specific tasks within a job or eliminates the job entirely, forcing workers to shift to other roles or industries
Labor augmentation	The use of AI or technology to enhance human workers' capabilities rather than replace them, making workers more productive or effective
Comparative advantage (labor)	The economic principle that workers should specialize in tasks they perform relatively better than alternatives — used to assess which human skills remain valuable even when AI is capable across many domains
Skill premium	The wage difference between workers with high-demand skills (e.g., technical or analytical) and those without; tends to increase as automation raises the value of skills AI cannot easily replicate
Structural unemployment	Long-term joblessness caused by a mismatch between workers' skills and the skills demanded by the labor market, often triggered by technological change

Term	Definition
Universal basic income (UBI)	A policy proposal in which the government provides all citizens with a regular, unconditional cash payment, often discussed as a response to large-scale job displacement from automation
Reskilling / upskilling	Reskilling: training workers to perform a different job or role. Upskilling: training workers to perform their current role at a higher level. Both are proposed responses to automation-driven displacement
Prompt engineer	A professional who designs, refines, and optimizes the text inputs (prompts) given to AI language models to produce desired outputs; one of the new job categories created by generative AI
AI auditor	A professional who evaluates AI systems for accuracy, fairness, bias, legal compliance, and ethical alignment; an emerging role in AI governance
Reinforcement learning from human feedback (RLHF)	A machine learning technique in which human evaluators rate AI outputs, and those ratings are used to train the AI to produce better responses; requires ongoing human labor to improve AI systems
Laborem Exercens	A 1981 papal encyclical by Pope John Paul II ("On Human Work") that articulates the Catholic Church's teaching on the dignity of work, the rights of workers, and the relationship between labor and capital
Dignity of work	The Catholic social teaching principle that work is not merely a means of economic production but an expression of human dignity — through work, people participate in God's creation and develop as persons
Priority of labor over capital	The Catholic social teaching principle, articulated in *Laborem Exercens*, that human labor must take precedence over capital and technology — tools exist to serve workers, not the other way around
Preferential option for the poor	The Catholic social teaching principle that moral and policy decisions should prioritize the needs of the most vulnerable members of society — applied to AI, this means ensuring displaced or low-skill workers are not left behind
Solidarity	The Catholic social teaching virtue of recognizing our mutual dependence and shared responsibility for one another's welfare — in the context of AI, it calls for collective action to ensure technological benefits are broadly shared
Common good	The Catholic social teaching principle that social, economic, and political arrangements should benefit all members of society, not just a privileged few — AI development should be evaluated by whether it serves the common good

10. Next Week!

Can AI be programmed to be ethical? This class will discuss moral dilemmas involving AI, such as its role in the military (autonomous drones) and medical decision-making (AI

doctors). We will examine real-world ethical debates, including the controversial topic of AI-powered weapons. A class debate will tackle whether AI should be allowed to make life-and-death decisions.

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